A Universal Quantum Computer From Relativistic Motion

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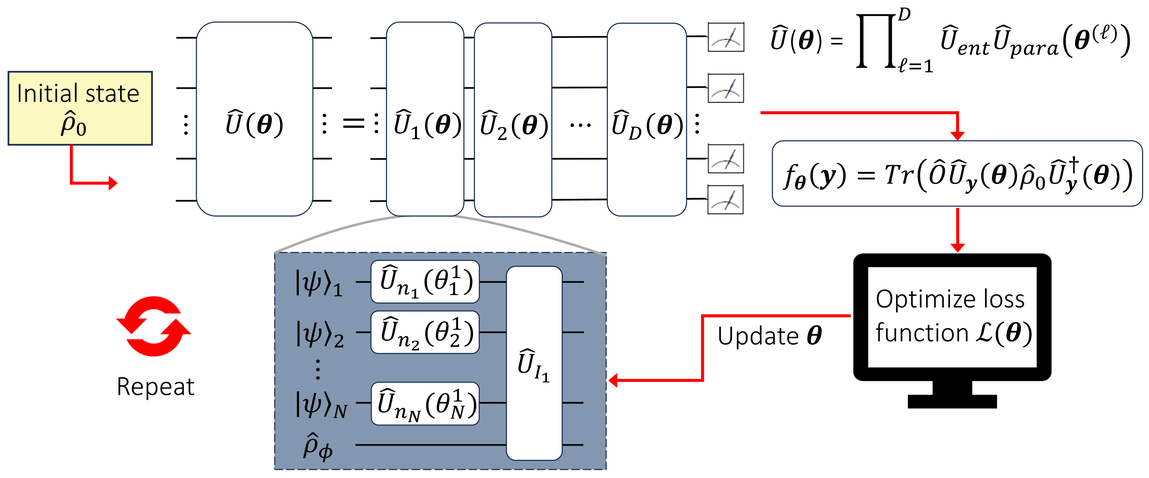
Basic Idea: A relativistic, universal quantum computing architecture using a variational quantum circuit approach consisting of single-qubit rotations parameterized by the proper time intervals of the qubits' trajectories through spacetime, and entangling gates mediated by a relativistic quantum field. A lower bound is also derived on the channel fidelity that shows the existence of parameter regimes in which all entangling operations are effectively unitary, despite the noise generated from the presence of a mediating quantum field.

Variational Quantum Circuits (VQC)



Our framework is naturally captured by VQCs, which work by tuning parameters in a parametrized quantum circuit according to the optimization of a loss function that takes circuit measurements as input. Another important consideration is that VQCs are also noise robust.

Local Operations

Consider array of UDW systems

Internal dynamics prescribed by free Hamiltonian

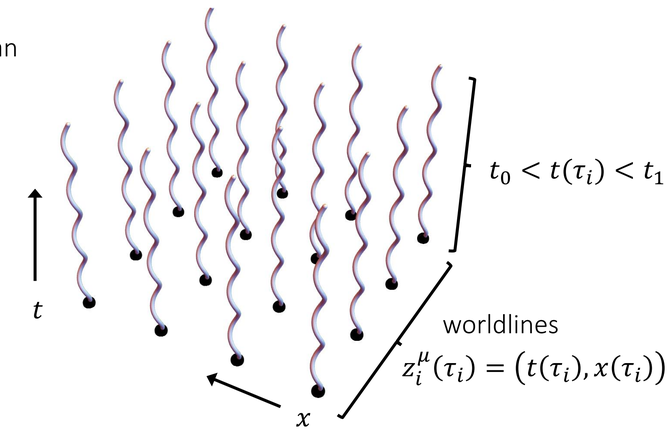
$$\hat{H}_i(\Omega_i, \vartheta_i, \varphi_i) = \frac{\Omega_i}{2}\boldsymbol{n}(\vartheta_i, \varphi_i)\cdot\hat{\boldsymbol{\sigma}}_i$$

where Ω_i are UDW system energy gaps.

→ 1-qubit rotation operators:

$$\hat{U}_{para}(\boldsymbol{\theta}^{(\ell)}) = \prod_{i=1}^{N} e^{-i\hat{H}_i(\Omega_i,\vartheta_i,\varphi_i)\Delta\tau_i}$$

with rotation angles $\theta_i = \Omega_i\Delta\tau_i/2$ and trainable parameters $\boldsymbol{\theta} = \{\theta_i, \vartheta_i, \varphi_i\}_i$



Rotation angles controlled through the proper time interval $\Delta\tau_i$ by changing spacetime trajectories of UDW systems rather than through their energy gaps.

e.g. circular trajectories in Minkowski: $\Delta\tau_i = \Delta t\sqrt{1 - R^2\omega_i^2}$

Entangling Operations

Consider UDW scalar field interaction Hamiltonian density

$$\hat{h}_i(\mathrm{x}) = \lambda\Lambda_i(\mathrm{x})\hat{\mu}_i\hat{\phi}(\mathrm{x})$$

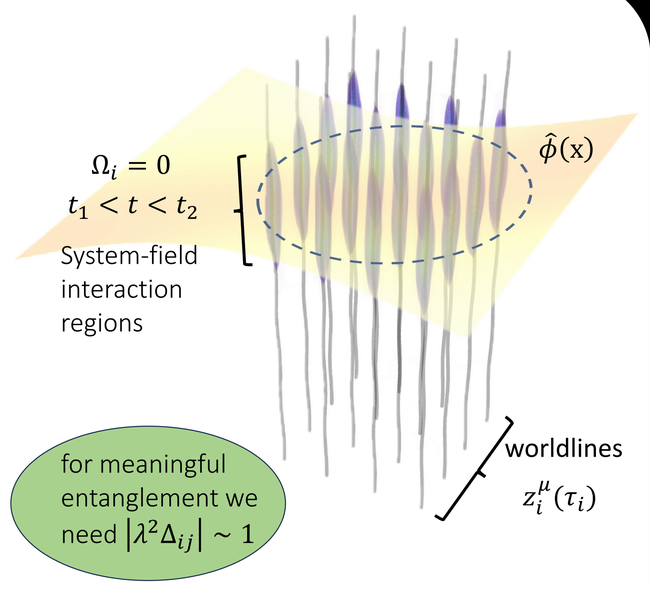
with coupling constant λ and spacetime smearing $\Lambda_i(\mathrm{x})$.

Time-evolution plus tracing out field results in decohering channel:

$$\mathcal{E}(\hat{\rho}_0) = \hat{U}_c\mathcal{E}_\phi(\hat{\rho}_0)\hat{U}_c^\dagger \quad \text{with} \quad \hat{U}_c = e^{-i\frac{\lambda^2}{2}\sum_{i<j}\Delta_{ij}\hat{\mu}_i\hat{\mu}_j}$$

Δ_{ij} are parameters that depend on retarded propagation of UDW system interaction regions through field.

UDW systems can couple to field simultaneously, so $\hat{U}_c$ is a globally entangling unitary on all systems.



Noise Analysis

How bad is the noise from interacting with the field?

Channel Fidelity: $F(\mathcal{E}(\hat{\rho}), \hat{U}_c\hat{\rho}\hat{U}_c^\dagger) \geq Tr(\hat{\rho}^2)e^{-2N^2\lambda^2\mathcal{W}}$

Arbitrary initial state

Field-dependent parameter determining local noise experienced by each system

Example: Gaussian smearing in Minkowski

$$\Lambda_i(\mathrm{x}) = \frac{1}{(2\pi\sigma^2)^{3/2}}e^{-\frac{(t-t_c)^2}{2T^2}-\frac{|x-x_i|^2}{2\sigma^2}}$$

$$\mathcal{W} = \frac{T^2}{4\pi(\sigma^2+T^2)}$$

$$\Delta_{ij} = \frac{T^2 e^{-\frac{L_{ij}^2}{4(\sigma^2+T^2)}}}{\sqrt{4\pi(\sigma^2+T^2)}L_{ij}}\mathrm{erf}\left(\frac{TL_{ij}}{2\sigma\sqrt{\sigma^2+T^2}}\right)$$

Where on the spectrum is our setting?

1) $|\lambda^2\Delta_{ij}| \sim 1$ — Meaningful entanglement
2) $N^2\lambda^2\mathcal{W} \ll 1$ — Negligible noise

→ $N^2\mathcal{W} \ll \Delta_{ij}$

True for general spacetimes

Δ_{ij} typically scales with interaction times

→ $L_{ij} \ll T$

$T \equiv$ interaction duration scale
$\sigma \equiv$ effective interaction size
$t_c \equiv$ interaction duration centre
$L_{ij} \equiv$ distance between systems

e.g. for $\lambda \sim 10^{-4}$ and $N = 100$ → Per layer: $F \geq 99.998\%$

100 layers: $F \geq 99.8\%$

Universality

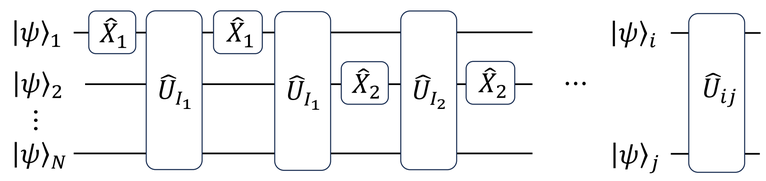
How do we know it is universal?

→ $\hat{U}_{para}(\boldsymbol{\theta})$ can implement arbitrary 1-qubit rotations and we have a 2-qubit entangling gate

BUT

$\hat{U}_c$ entangles everything, which is not known to be universal.

Can consider $\mathcal{O}(N)$ products of 2 RVQC layers with Pauli X rotations to distill down to entangling gate between only 2 systems



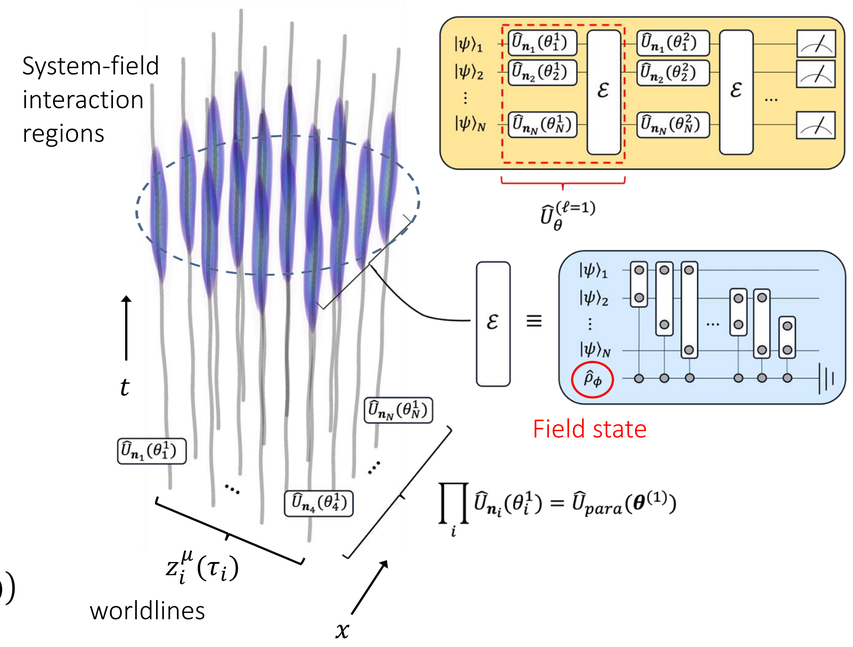
→ Now we have universality!

Relativistic VQC

The setup works as follows:

1. Apply single qubit rotations $\hat{U}_{para}(\theta_i)$ to each system
2. Couple all systems to field and entangle them through channel $\mathcal{E}$
3. Repeat steps 1 and 2 D times, then measure output with respect to $\hat{O}$:

$$f_{\boldsymbol{\theta}}(\boldsymbol{y}) = Tr(\hat{O}\hat{U}_{\boldsymbol{y}}(\boldsymbol{\theta})\hat{\rho}_0\hat{U}^\dagger_{\boldsymbol{y}}(\boldsymbol{\theta}))$$

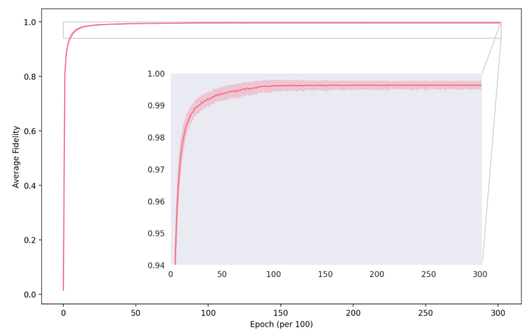


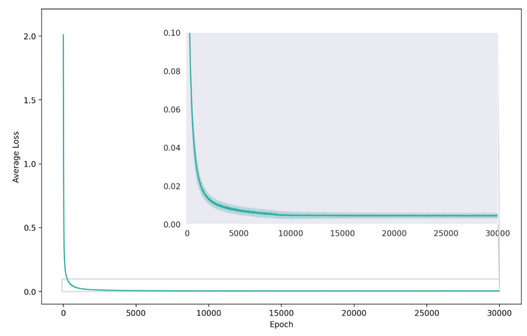
Key distinctions between typical non-relativistic setups and ours:

1. Trainable parameters explicitly involve spacetime motion of detectors through their proper time during evolution
2. Interactions between detectors are entirely mediated through local interactions with quantum field
3. Presence of noise terms from entanglement between detectors and quantum field

Example: Quantum Fourier Transform

As an illustration of the relativistic VQC, we consider the quantum Fourier transform for $N = 6$ qubits. We fix the qubit geometry by arranging them into a 3×2 square lattice with separations of $(\Delta x, \Delta y) = T\lambda^2(5, 3)$, where $T = 10^8$ and $\lambda = 10^{-4}$. Then we set the spatial smearing width to $\sigma = 10^{-8}$ and fix the entangling gate operators $\hat{\mu}_i = \hat{\sigma}_z$ for each qubit. We consider $D = 50$ layers and find that we are able to approximate the quantum Fourier transform $\hat{U}_{QFT}$ with an average final precision in the scaled Hilbert-Schmidt norm squared of $\frac{1}{N}\|\hat{U}_\theta - \hat{U}_{QFT}\|^2_{HS} = 0.00444 \pm 0.00138$ and an average final fidelity of $F = 0.9962 \pm 0.0014$ between $\hat{U}_\theta$ and $\hat{U}_{QFT}$ acted on the same ensemble of randomly-generated states.





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